CTI-4

Silicon Carbide Inverter

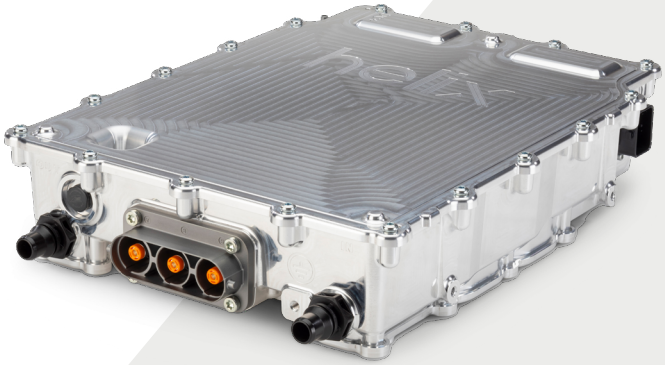
CTI-4 is the next generation in 3 Phase Silicon Carbide Inverter technology from Helix. Designed specifically for hybrid and electric vehicles, this cutting-edge inverter combines advanced technologies with Hardware and Software necessary to meet the rigors of the required applications.

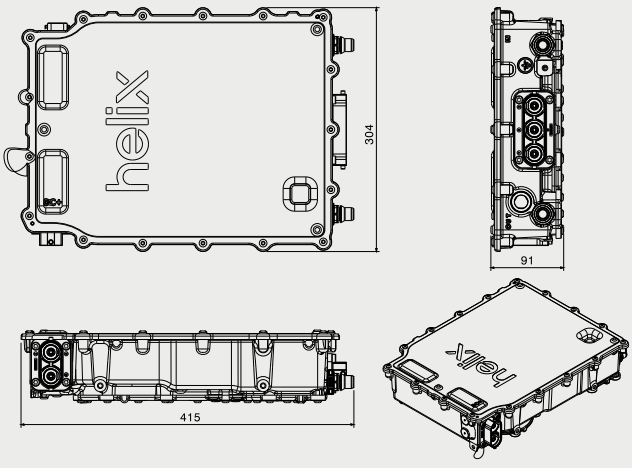
Featuring high DC bus voltage capability and state-of-the-art liquid cooling technology, the CTI-4 allows for exceptional power, phase current, and switching frequency in a remarkably compact and lightweight unit. Supporting both surface and internal permanent magnet synchronous motors, the CTI-4 ensures optimal performance and versatility.

This self-contained inverter is intended to be placed adjacent to a high voltage battery and connected to the electric drive unit via phase cables. Engineered and manufactured in-house, custom variations can be tailor-made to meet specific client requirements in terms of performance and packaging.

The CTI-4 offers full synchronous closed-loop motor control with torque control and speed control functions available through a CAN interface. The control algorithms incorporate standard motor control features such as Space Vector Modulation, as well as advanced features for Over-Modulation, Discontinuous Switching and Optimised Switching Pattern.

With a proven track record in high-performance road car applications and high-profile motorsports, including the Lotus Evija, VW ID-R Pikes Peak and Nürburgring Record Holder, the CTI-4 leverages Helix's extensive experience in silicon carbide (SiC) technology and Functional Safety. It empowers electric motors to achieve peak power of over 550kW and continuous power of 300kW. Primarily designed for automotive applications, the CTI-4 is also compatible with Helix's high-speed Scalable Core technology motors.





Key features

- SiC technology for ultimate compactness and efficiency
- Extremely compact & robust (7.4l volume)
- Simple installation and integration (direct motor mounting available)
- Peak power > 550kW
- Peak phase current > 750 Arms
- Peak power density >52.5kW/kg
- Low mass 10.5kg
- AUTOSAR 4.4
- ISO14229 UDS functionality
- ISO11898 CAN interfaces available
- Variable PWM up to 40kHz
- DC Bus voltage ripple reduction via high capacitance and control strategies
- Ingress Protection: IP6K9K

Powertrain Advantage

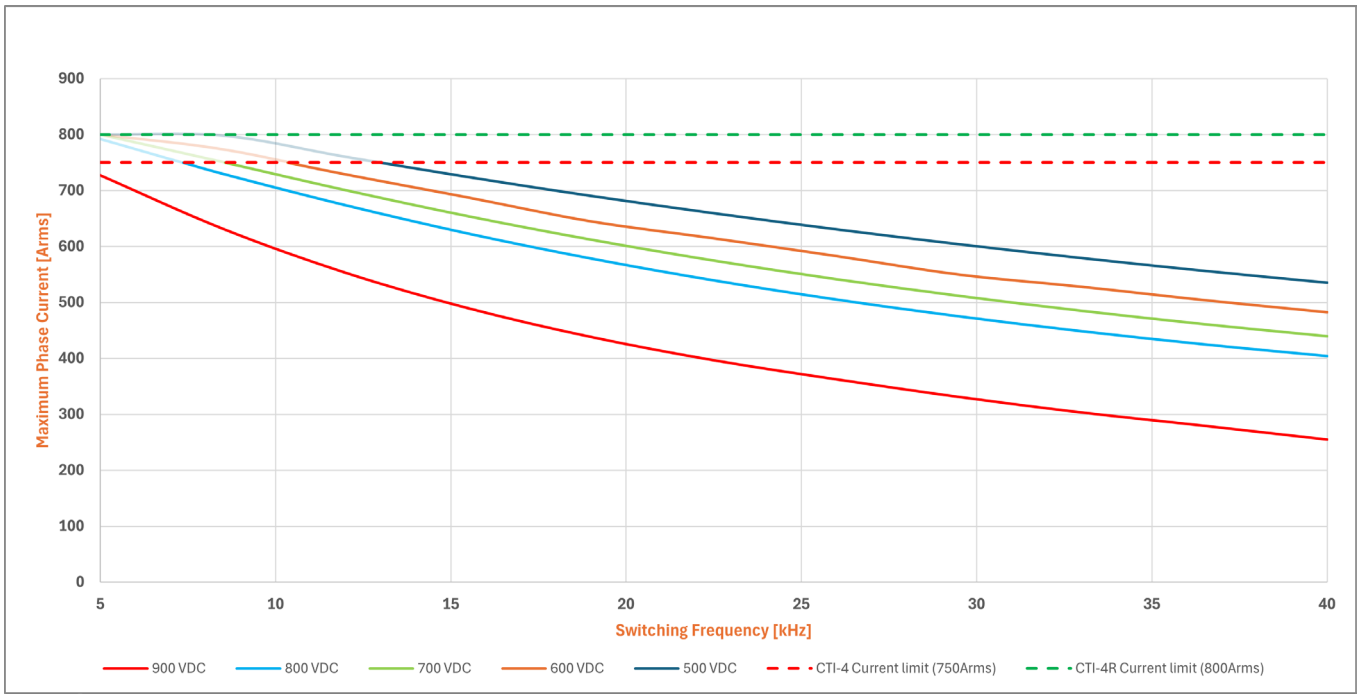
CTI-4 Technical Data

Technical Data	Unit	CTI-4	CTI-4R
Part Number	P/N	2BA03	2BA04
Inverter Type	-	3 Phase Silicon Carbide Inverter	
Applications	-	Automotive, Passenger Car, Commercial Vehicle (class 1-8), Off-Highway, Rail, Marine, Motorsport, Power Generation, Motorcycle, Hybrid, FCEV, REEV and BEV applications	
Physical Specifications			
Casing Width	mm	304	
Casing Length	mm	415	
Casing Height	mm	91	
Mass[1]	kg	10.5	
Performance Specifications			
Peak Power[4]	kW	550	588
Continuous Power	kW	300	
Maximum Phase Current	Arms	750	800
Continuous Phase Current[3]	Arms	435	
Continuous DC Current[3]	A	500	
Mechanical Specifications			
Maximum Linear Vibration (normal operation)	-	ISO16750-3:2023	
Impact Condition	-	15 g	
Environmental/Cooling Specifications			
Cooling Medium	-	50% Ethylene Glycol	
Coolant Inlet Temperature	°C	-40 to +65	
Typical Coolant Flowrate	l/min	10-20	
Coolant Inlet Pressure	bar (gauge)	1.0-3.0	
Minimum/Maximum Inverter surrounding temperature (Installed)	°C	-40 to +85	
Ingress Protection Standard	-	IP67 / IP6K9K	
Maximum Altitude	m	5000	
Coolant Quality	µm	250	
Electrical Specifications			
Maximum DC Voltage (Operating)	VDC	925	
Minimum DC Voltage[2]	VDC	500	
DC Capacitance	µF	360	
Y Capacitance	nF	200	
Peak Inverter Efficiency	%	> 99.0% @ 200kW	
Minimum Phase/DC to Ground Resistance	Mohm	UN Reg10	
PWM Frequency	KHz	Up to 40	
CAN Termination	-	CAN termination not fitted	
Communication	-	2 CAN communication channels ISO 11898-2	
Software Specifications			
AUTOSAR version	-	AUTOSAR 4.4.0	
Storage Specifications			
Minimum/ Maximum Storage Temperature	°C	-40 to +85	

NOTES:

1. Mass Including: phase cable glands, coolant hose fittings, sensor glands
2. Performance derating will occur depending on motor specification, battery voltage and temperatures
3. 70mm^2 Coroplast cable <1M in length, 1000Hz electrical frequency, 50°C coolant, 25°C ambient for 10 minutes @600V
4. 800V, 0.75pf assumed

CTI-4 Inverter Peak Phase Current Performance Characteristics

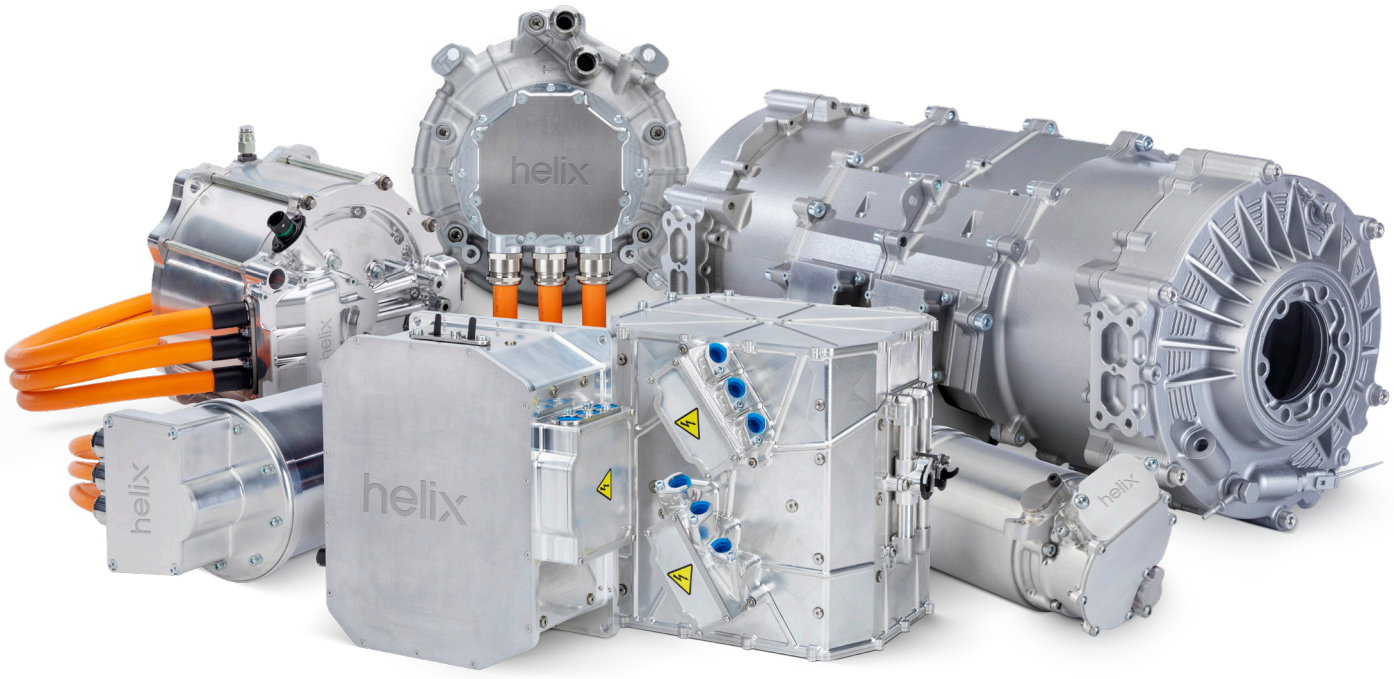


NOTES:

Coolant Temp = 50°C
Inverter Coolant Flow Rate = 10 Litres/min

The performance ratings and graphs are based on typical operating conditions as noted. There are several variables that may change the motor performance, including coolant flow rate, operating temperature, inverter settings and parameters, etc. For actual performance, the motor must be evaluated in its final system and application.

All data subject to change without notice.



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Powertrain Advantage

+44 (0)1908 278600 contact@ehelix.com ehelix.com

Helix, Merlewood Drive, Shenley Wood, Milton Keynes MK5 6GR

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